Domain Case Study 2: Prosumer Media Asset Governance & Quality Heuristics

Document Classification: [SIMULATED BA CASE STUDY — DERIVED FROM PROJECT DOMAIN]

Domain: Prosumer Digital Asset Management, Commercial Photography, Media Workflows

Target Scenario: A boutique wedding and commercial photography studio managing 12 TB of high-resolution RAW and JPEG imagery across local NVMe storage and Google Drive.

1. Case Study Overview & Business Problem

Business Context

"Lumina Photo Studios" shoots 60 commercial events per year. Each event generates 3,000 to 5,000 camera exposures shot in dual-slot mode (Slot 1: Sony `.ARW` RAW 45 MB; Slot 2: High-res JPEG 12 MB). Across editing, culling, client proofing, and social media exports, redundant versions of the same shots proliferate across folders (`Raw_Imports/`, `Cull_Round1/`, `Client_Web/`, `Final_Print/`).

The Problem

The studio's 4 TB editing workstation SSDs were permanently full (> 95%), causing editing software (Adobe Lightroom / DaVinci Resolve) cache crashes. Furthermore, Google Drive storage tiers escalated from $9.99/mo to $49.99/mo per photographer. Manual file culling consumed an estimated **8 to 12 hours per week** of high-cost senior photographer time.

2. Stakeholder Analysis

Stakeholder	Role	Primary Goal	Core Concern
Studio Principal / Lead Photographer	Business Owner / User	Reclaim local SSD speed and slash Google Drive subscription tiers.	Accidentally trashing pristine 45 MB RAW files or unedited camera originals.

Stakeholder	Role	Primary Goal	Core Concern
Associate Retoucher	Production User	Fast, visual side-by-side inspection before marking files for removal.	Sluggish software with poor thumbnail previews or complex command lines.
Technical Business Analyst	Requirements Lead	Model "Keep Best Quality" heuristics, drag-and-drop ergonomics, and undo mechanics.	Edge cases where edited JPEGs have different crop dimensions than original RAW files.

3. As-Is vs. To-Be Process & Gap Analysis

MERMAID DIAGRAM

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flowchart LR
    subgraph "As-Is Painful Workflow"
        A[Photoshoot finishes: 4,000 photos] --> B[Photographer spends 10 hours manually opening photos]
        B --> C[Manual guess: Which file is the master?]
        C --> D[Accidental deletion of RAW camera file]
        D --> E[Client demands print revision -> Disaster]
    end

    subgraph "To-Be Duplicate Bin Workflow"
        F[Drag folder onto DropZoneWidget] --> G[Intelligent Detection: O(n) Size Pre-Filter]
        G --> H[Selection Strategy: KEEP_BEST_QUALITY]
        H --> I[System auto-keeps 45MB RAW, marks 12MB JPEG duplicate]
        I --> J[Side-by-side preview verifies identical EXIF timestamps]
        J --> K[Safe Archive: 180 GB recovered in 45 seconds; Ctrl+Z active]
    end
```

4. Key Functional Requirements (FRD Summary)

- **FR-MEDIA-01 (Quality Hierarchy Rule):** When executing `KEEP_BEST_QUALITY`, uncompressed RAW formats (`.arw`, `.cr2`, `.nef`, `.dng`) shall rank above lossless formats (`.tiff`, `.png`), which shall rank above lossy formats (`.jpg`, `.webp`).
- **FR-MEDIA-02 (EXIF Metadata Validation):** The system shall extract and compare EXIF timestamps

- **FR-MEDIA-02 (EXIF Metadata Validation):** The system shall extract and compare EXIF timestamps (`DateTimeOriginal`), camera model, and lens serial number to differentiate identical burst shots from exact file duplicates.
- **FR-MEDIA-03 (Side-by-Side Thumbnail Rendering):** The GUI preview widget shall render side-by-side visual image comparisons with zoom capability, bounded by an LRU cache of 200 thumbnails to protect RAM.
- **FR-MEDIA-04 (Drag-and-Drop Ingestion):** Users must be able to drag folders directly from macOS Finder or Windows Explorer onto the window to begin scanning.

5. Agile User Story & Gherkin Acceptance Criteria

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Feature: Automated High-Quality Media Preservation
  As a commercial photographer
  I want the system to automatically select compressed web exports for removal while preserving original R
  So that I can clean up client delivery folders without risking high-resolution master assets.

  Scenario: Automated selection of RAW over JPEG duplicate
    Given a duplicate group with "wedding_041.arw" (RAW, 48 MB) and "wedding_041.jpg" (JPEG, 6 MB)
    When I select the "KEEP_BEST_QUALITY" strategy
    Then "wedding_041.arw" must be preserved as the master file
    And "wedding_041.jpg" must be marked for safe archiving
    And the preview pane must display camera EXIF metadata confirming matching shutter timestamp.
```

6. Business Outcomes & Measurable Impact

- **Storage Reclaimed:** Reclaimed **1.8 TB** of redundant JPEG exports across 3 editing workstations.
- **Cloud Spend Reduction:** Downgraded 3 Google Workspace accounts, saving **$1,440 annually**.
- **Labor Efficiency:** Reduced weekly culling time from **10 hours down to 15 minutes** per event.
- **Data Protection:** Zero loss of original camera RAW files across 60 completed client projects.

7. Interview Preparation: How to Defend This Case Study

1. **How did you handle user skepticism regarding automated image deletion?**

   "Photographers are terrified of automated tools deleting their art. I addressed this with two critical BA

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design choices: first, the Side-by-Side Preview Pane with EXIF metadata comparison (showing exact shutter speed, aperture, and camera serial number); second, the Safe Archive Quarantine with instant `Ctrl+Z` *undo. Users tested it on sample folders and immediately trusted the platform."*

2. **What technical constraint did you uncover during requirements definition?**

   "We discovered that loading 45 MB RAW photos into memory for thumbnail previews rapidly caused Out-Of-Memory crashes in Qt. I formulated Non-Functional Requirement NFR-003, mandating an LRU Cache bounded at 200 pixmaps, and using Pillow thumbnail decoders rather than loading full-resolution sensor bitmaps into RAM."